

Problem Set 2 – Set identities

Session 2 · Sets & set operations

Part A · Notation and computation

1 NOTATION

Let $U = \{1, 2, \dots, 12\}$, $A = \{x \in U : x \text{ is even}\}$ and $B = \{x \in U : x \text{ is a multiple of } 3\}$.

- Write A and B in roster notation.
- Compute $A \cap B$, $A \cup B$, $A \setminus B$, $A \triangle B$ and B^c .

SOLUTION (a) $A = \{2, 4, 6, 8, 10, 12\}$, $B = \{3, 6, 9, 12\}$.
 (b) $A \cap B = \{6, 12\}$; $A \cup B = \{2, 3, 4, 6, 8, 9, 10, 12\}$; $A \setminus B = \{2, 4, 8, 10\}$;
 $A \triangle B = \{2, 3, 4, 8, 9, 10\}$; $B^c = \{1, 2, 4, 5, 7, 8, 10, 11\}$.

2 $\in$ VS $\subseteq$

Let $C = \{1, \{1\}, \{1, 2\}, \emptyset\}$. Decide true or false, and in each case quote the definition that settles it.

- $1 \in C$
- $\{1\} \in C$
- $\{1\} \subseteq C$
- $\{1, 2\} \subseteq C$
- $\emptyset \subseteq C$
- $|C| = 4$

SOLUTION (a) True – 1 is listed. (b) True – $\{1\}$ is listed as an element. (c) True – the only element of $\{1\}$ is 1, and $1 \in C$. (d) False – it would need $2 \in C$, and 2 is not an element of C . (e) True – vacuously, for every set. (f) True.

3 POWER SET

Let $A = \{a, b, c\}$.

- List $\mathcal{P}(A)$ in full, organised by the size of the subset.
- State $|\mathcal{P}(A)|$ and explain in one sentence why $|\mathcal{P}(A)| = 2^{|A|}$ for any finite A .

SOLUTION (a) $\emptyset$; $\{a\}$, $\{b\}$, $\{c\}$; $\{a, b\}$, $\{a, c\}$, $\{b, c\}$; $\{a, b, c\}$.
 (b) $|\mathcal{P}(A)| = 8 = 2^3$. Building a subset means deciding, independently for each of the n elements, whether it is in or out – two choices n times, so 2^n subsets.

Part B · Six identities to prove

4 DISTRIBUTIVITY

Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \cup (B \cap C)$ be arbitrary. Then $x \in A$, or $x \in B \cap C$. If $x \in A$ then x is in both $A \cup B$ and $A \cup C$, so in their intersection. If $x \in B \cap C$ then $x \in B$ and $x \in C$, so again $x \in A \cup B$ and $x \in A \cup C$. Either way $x \in (A \cup B) \cap (A \cup C)$. ($\supseteq$) Let $x \in (A \cup B) \cap (A \cup C)$ be arbitrary, so $x \in A \cup B$ and $x \in A \cup C$. If $x \in A$ we are done. If not, then from the first $x \in B$ and from the second $x \in C$, so $x \in B \cap C$. Either way $x \in A \cup (B \cap C)$. ■

5 DE MORGAN

Let $A, B \subseteq U$. Prove that $(A \cap B)^c = A^c \cup B^c$.

SOLUTION ($\subseteq$) Let $x \in (A \cap B)^c$ be arbitrary. Then $x \in U$ and $x \notin A \cap B$, i.e. it is not the case that both $x \in A$ and $x \in B$. So $x \notin A$ or $x \notin B$, giving $x \in A^c$ or $x \in B^c$, hence $x \in A^c \cup B^c$. ($\supseteq$) Let $x \in A^c \cup B^c$ be arbitrary. Then $x \notin A$ or $x \notin B$, so it is not the case that x lies in both, i.e. $x \notin A \cap B$. As $x \in U$, $x \in (A \cap B)^c$. ■

6 DIFFERENCE

Prove that $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ for all sets A, B, C .

SOLUTION ($\subseteq$) Let $x \in A \setminus (B \cup C)$ be arbitrary. Then $x \in A$ and $x \notin B \cup C$, so $x \notin B$ and $x \notin C$. Hence $x \in A \setminus B$ and $x \in A \setminus C$, so x is in the intersection. ($\supseteq$) Let $x \in (A \setminus B) \cap (A \setminus C)$ be arbitrary. Then $x \in A$, $x \notin B$ and $x \notin C$, so $x \notin B \cup C$, giving $x \in A \setminus (B \cup C)$. ■

7 SYMMETRIC DIFFERENCE

Prove that $A \triangle B = (A \cup B) \setminus (A \cap B)$ for all sets A, B .

SOLUTION Recall $A \triangle B = (A \setminus B) \cup (B \setminus A)$. ($\subseteq$) Let $x \in A \triangle B$ be arbitrary, so $x \in A \setminus B$ or $x \in B \setminus A$. In the first case $x \in A \subseteq A \cup B$ and $x \notin B$, so $x \notin A \cap B$. The second case is symmetric. Either way $x \in (A \cup B) \setminus (A \cap B)$. ($\supseteq$) Let $x \in (A \cup B) \setminus (A \cap B)$ be arbitrary. Then x is in A or in B , but not in both. If $x \in A$ then $x \notin B$, so $x \in A \setminus B$; otherwise $x \in B$ and $x \notin A$, so $x \in B \setminus A$. Either way $x \in A \triangle B$. ■

8 SUBSETS

Prove that for all sets A, B : $A \subseteq B \iff A \cup B = B$.

SOLUTION $(\Rightarrow)$ Suppose $A \subseteq B$. Certainly $B \subseteq A \cup B$. Conversely, let $x \in A \cup B$ be arbitrary: if $x \in B$ we are done, and if $x \in A$ then $x \in B$ by the hypothesis. So $A \cup B \subseteq B$, and the two sets are equal.
 $(\Leftarrow)$ Suppose $A \cup B = B$. Let $x \in A$ be arbitrary. Then $x \in A \cup B$, which equals B , so $x \in B$. Hence $A \subseteq B$. ■

9 CARTESIAN PRODUCT

Prove that $A \times (B \cap C) = (A \times B) \cap (A \times C)$ for all sets A, B, C .

SOLUTION Elements here are ordered pairs, so an arbitrary element is written (x, y) .
 $(\subseteq)$ Let $(x, y) \in A \times (B \cap C)$ be arbitrary. Then $x \in A$ and $y \in B \cap C$, so $y \in B$ and $y \in C$. Hence $(x, y) \in A \times B$ and $(x, y) \in A \times C$.
 $(\supseteq)$ Let $(x, y) \in (A \times B) \cap (A \times C)$ be arbitrary. Then $x \in A$, $y \in B$ and $y \in C$, so $y \in B \cap C$ and $(x, y) \in A \times (B \cap C)$. ■